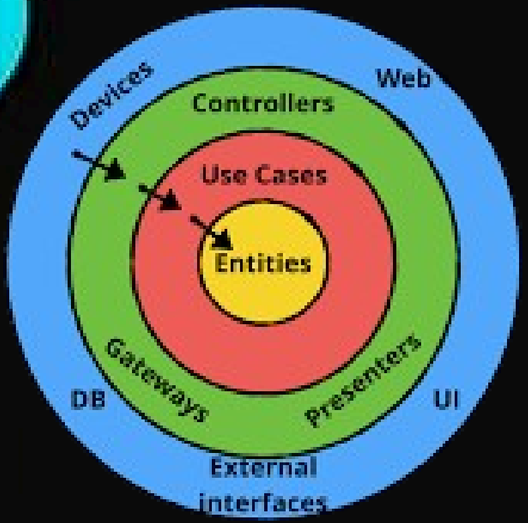
Building Ecommerce using Microservices



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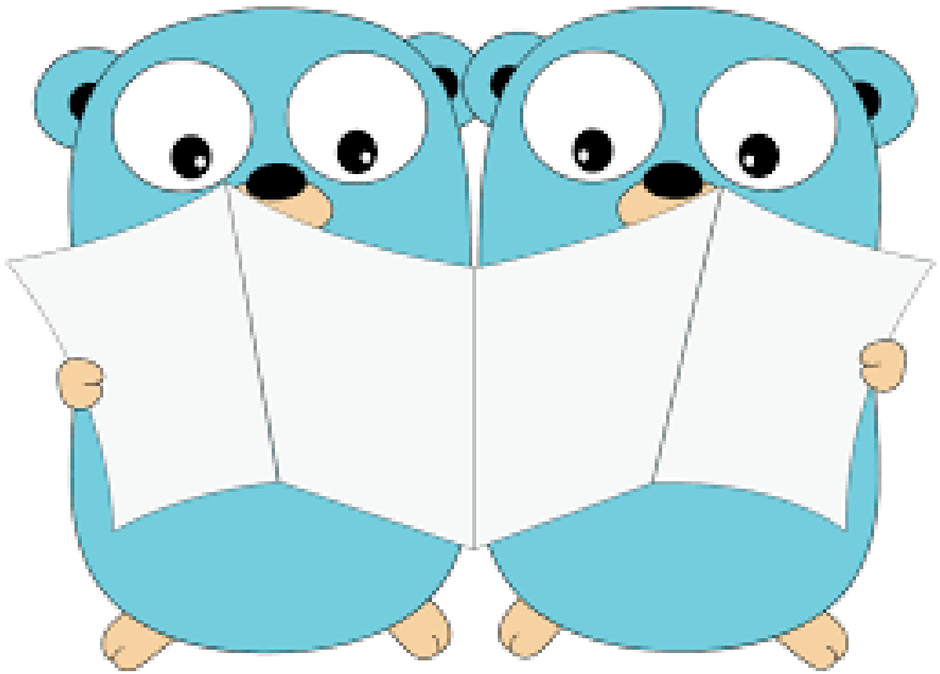


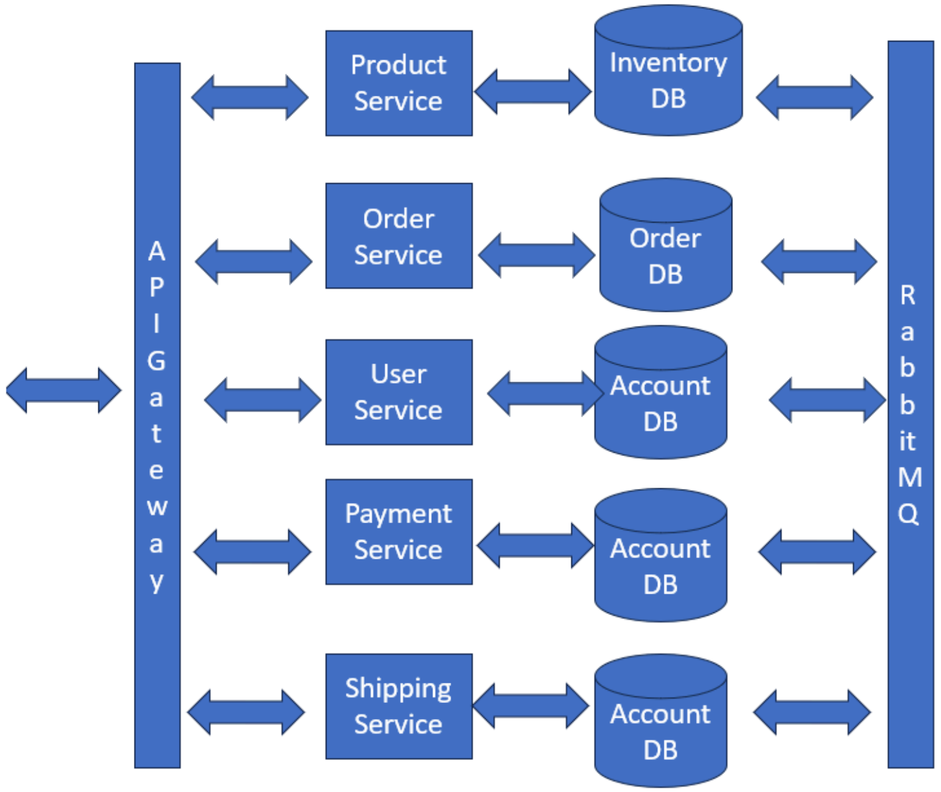
microservice architecture

A microservice is a independent compoent of a business logic. mostly is a breakdown of a monolith application into individual componetns that can scale and make the service fault tolerant

There are many advantages of microservces and they cater to different business needs as per design an architecture.

They can be :-

- Event driven
- Domain driven
- with webworker



key services

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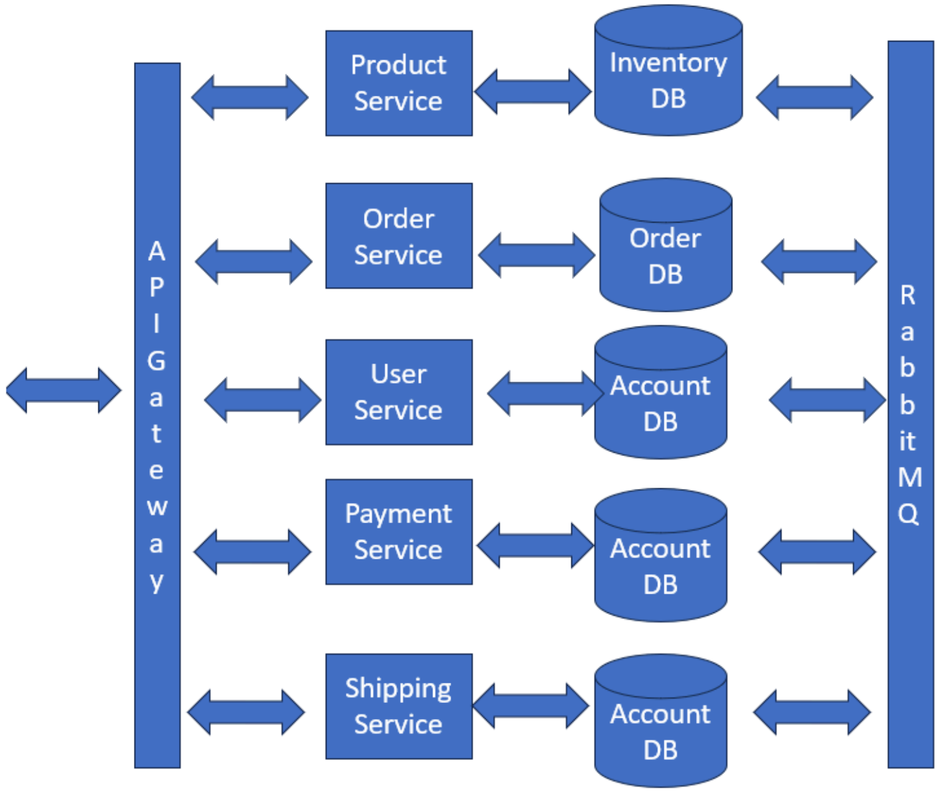

api gateway

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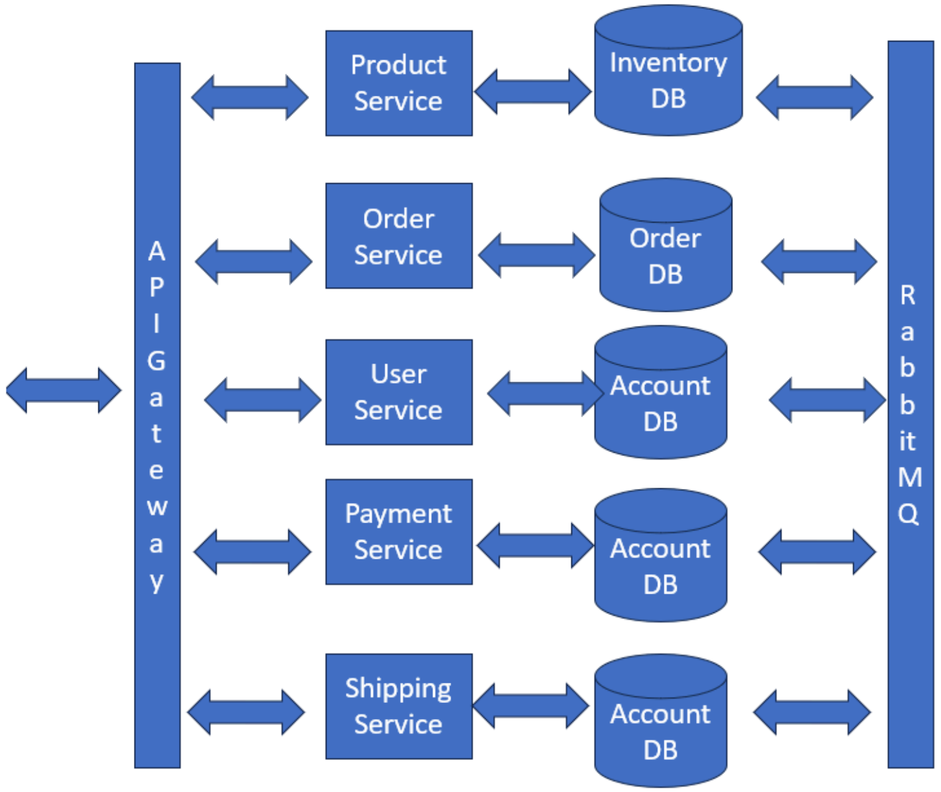

inter service communication

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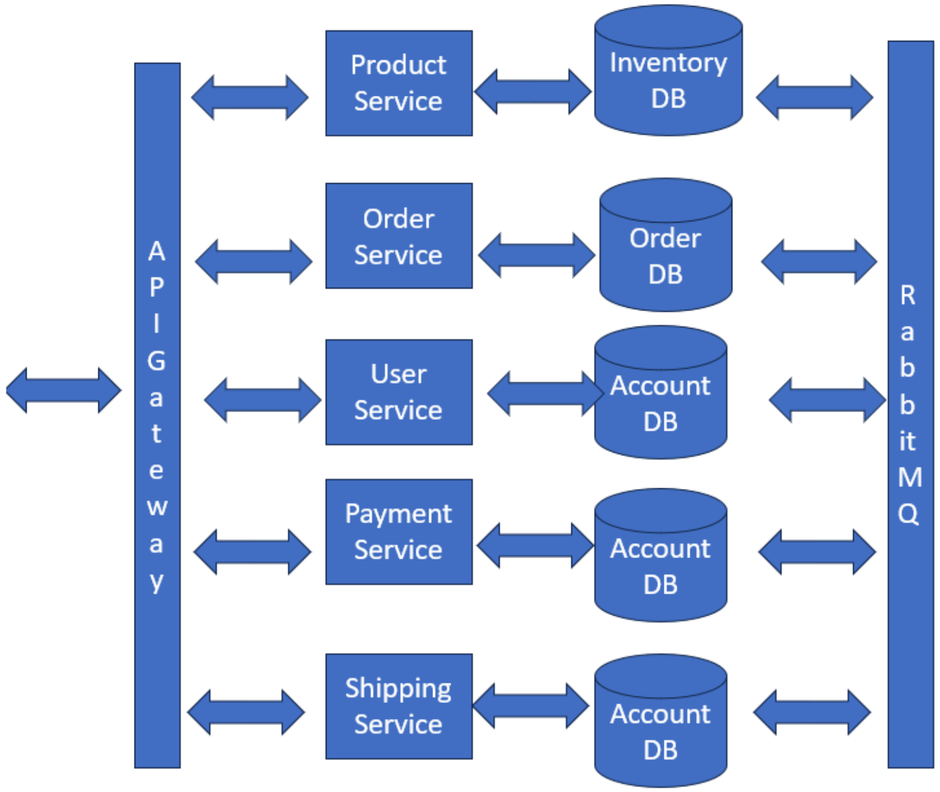

service discovery and configs

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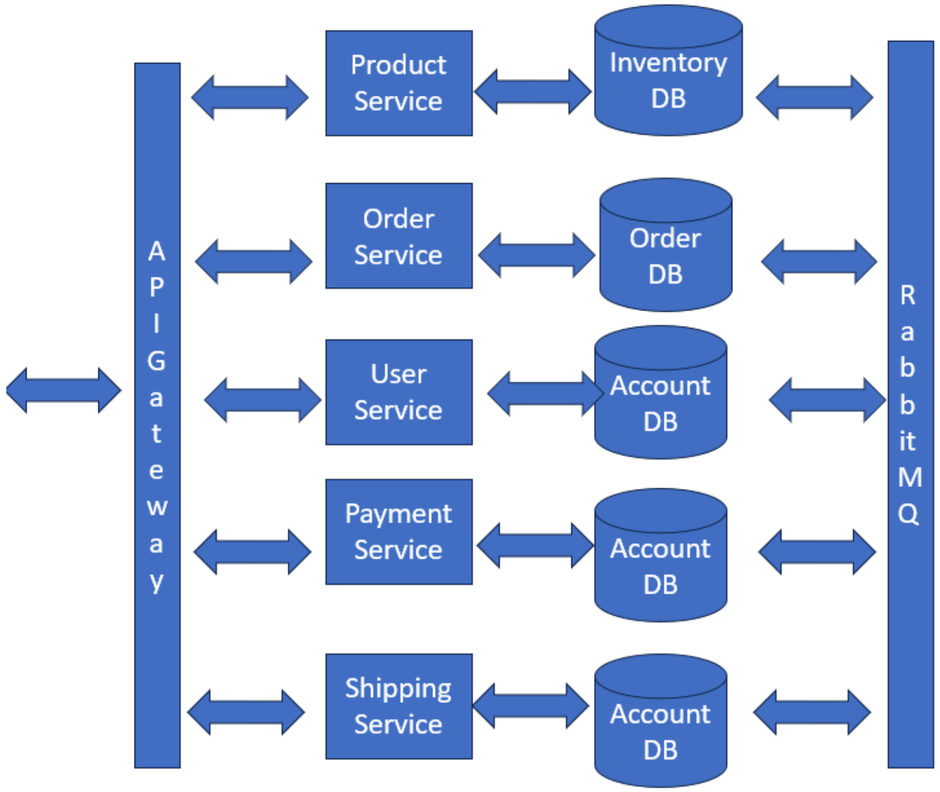

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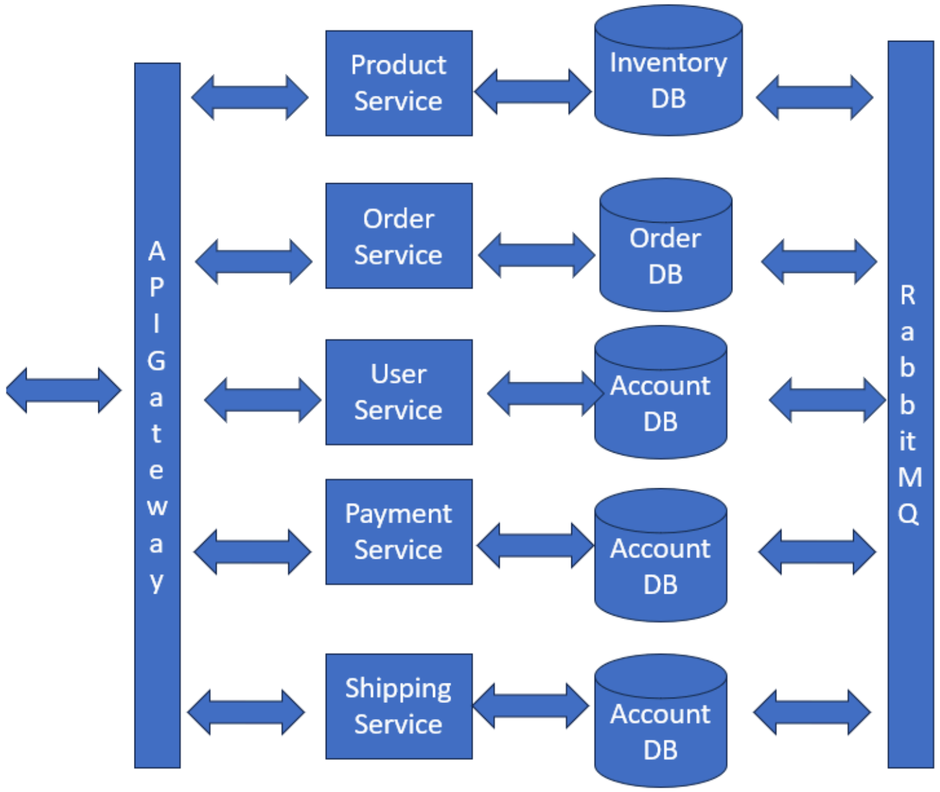

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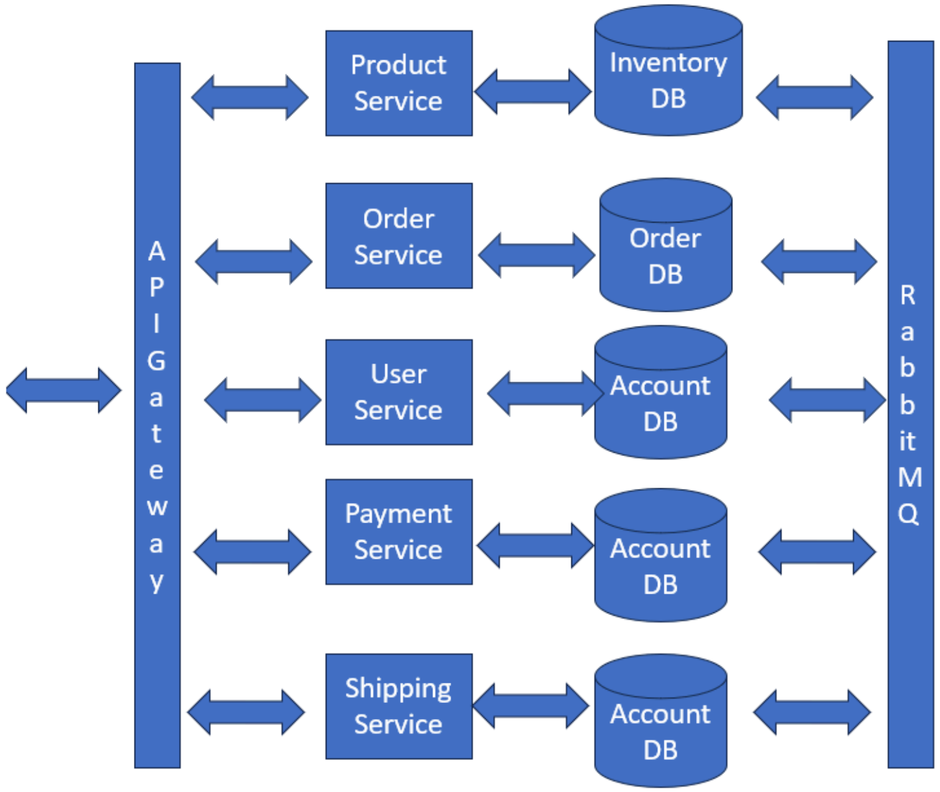

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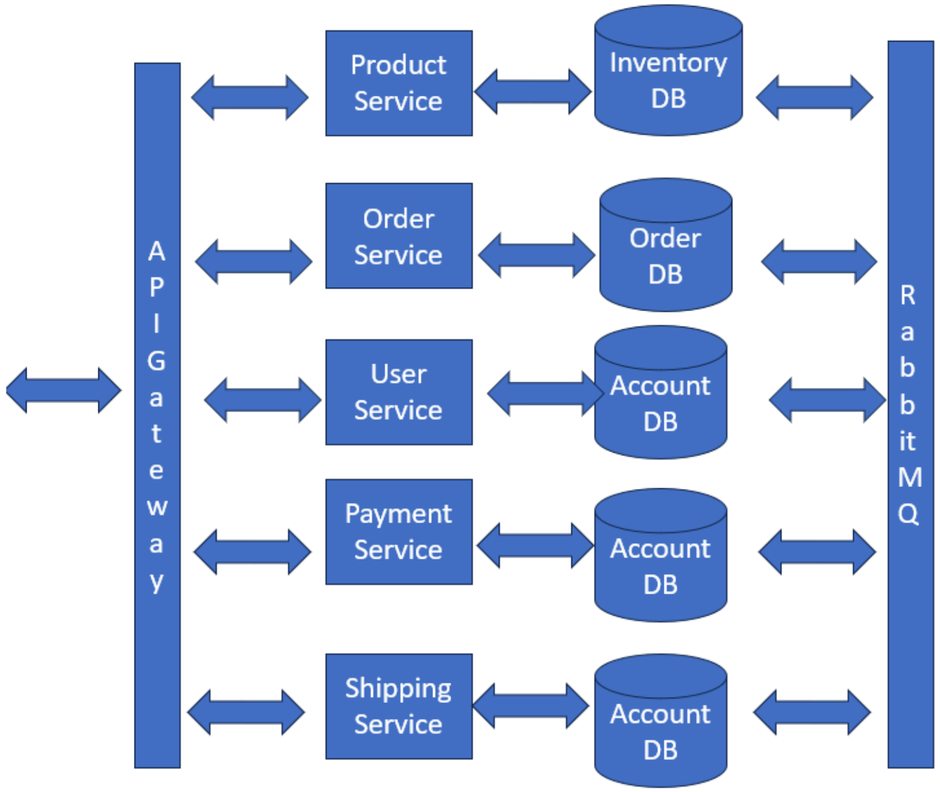

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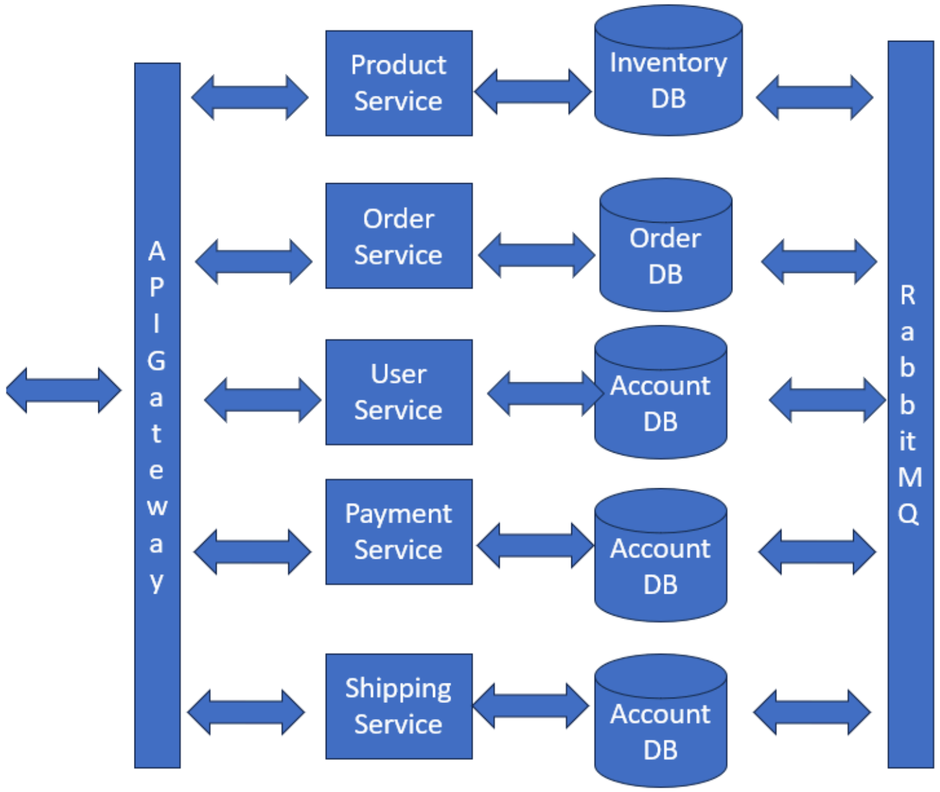
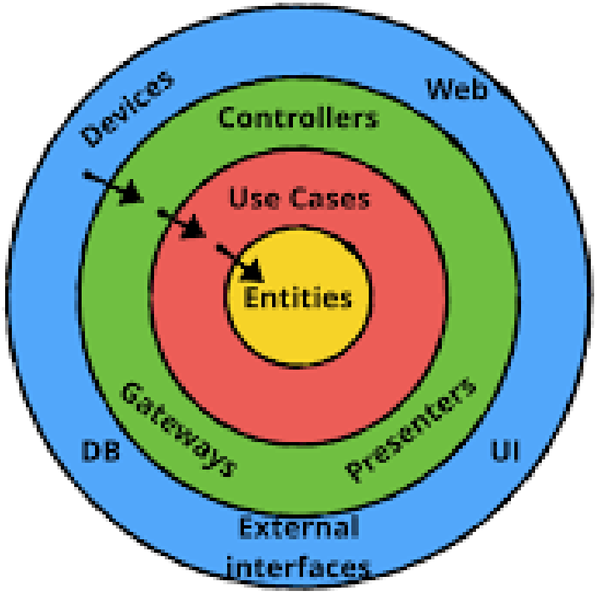


What is Domain Driven Design

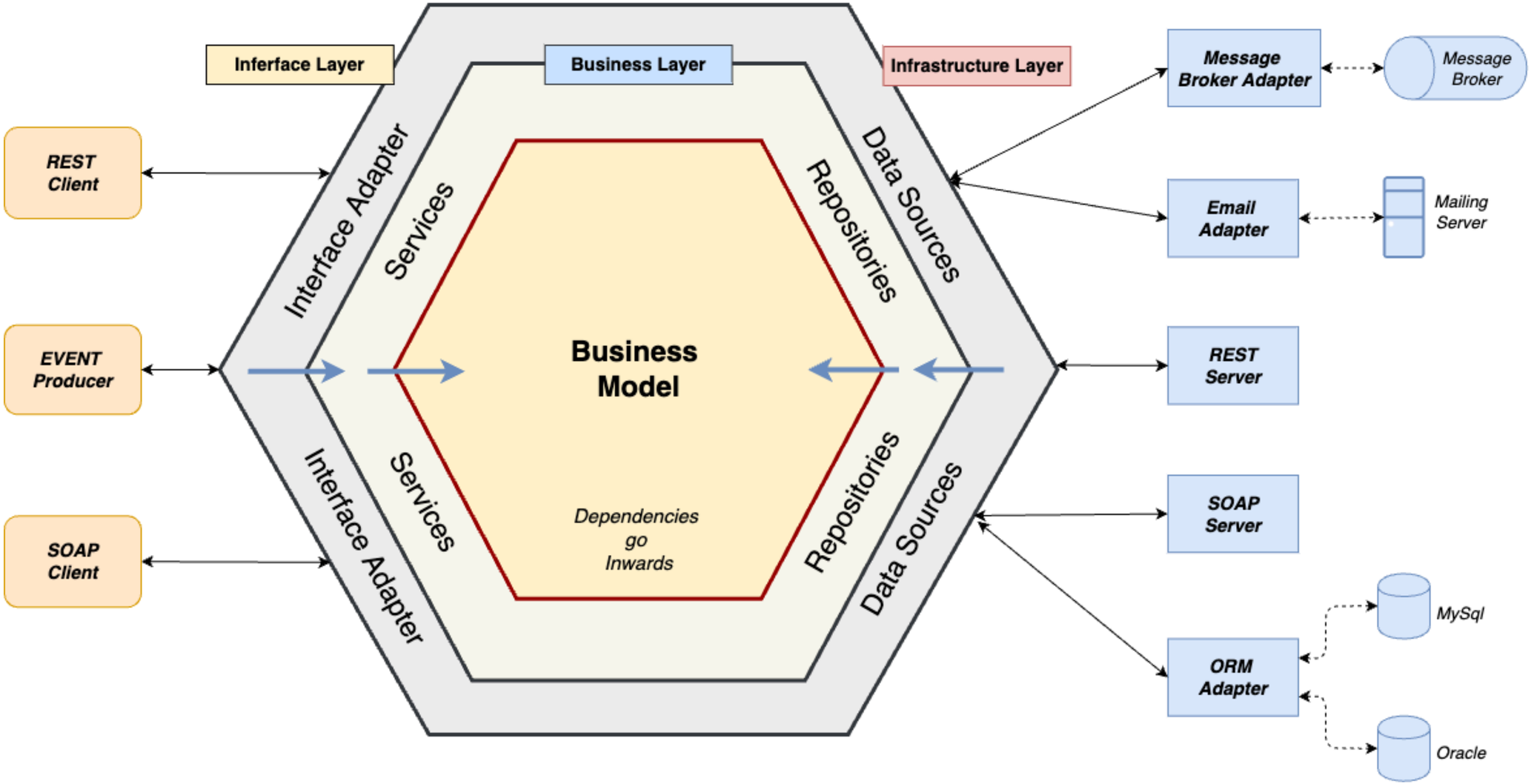
Domain Drivern Design is a software design pattern that saperates businesss logic from language infrastruture and other technical components. it has businss logic domain at its core and all other things that changes are wrapped like onion onto it.

it is important as the core business logic never changes but the outlying technologies changes but doesnot affect the business logic, It is like a port and adapter pattern where technologies like databases or message queues can be replaced easily without affecting anything.



What is Domain Driven Design

Hexagonal Architecture

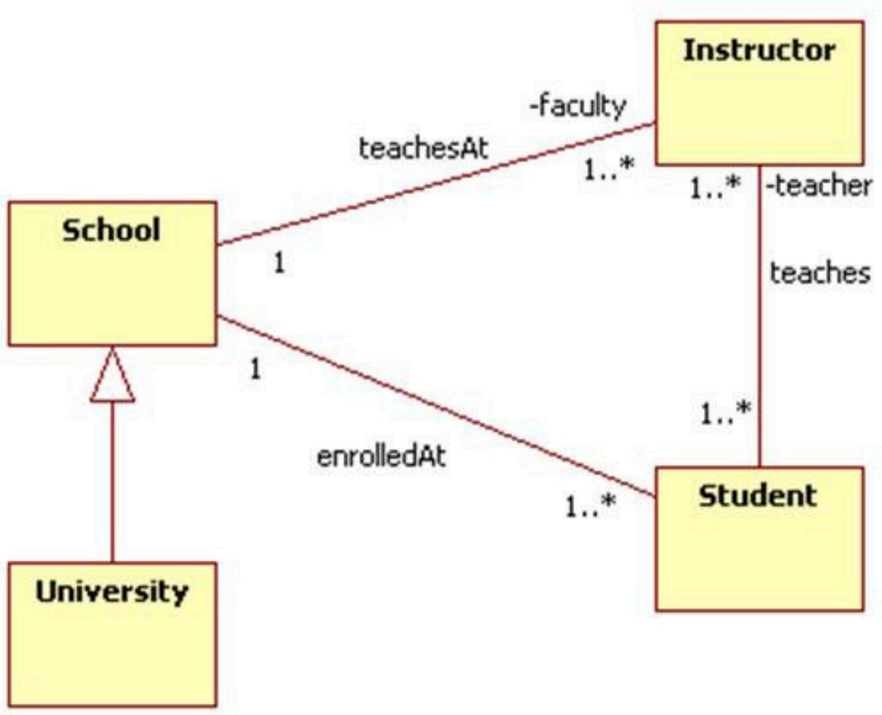


Domain Layer

The main business layer of the application that contains business models , methods, events and their working.

These are independent of any database or message queue or router and they mostly dont change.

they have classes or structs that defines objects or models and their methods and their interface signatures .

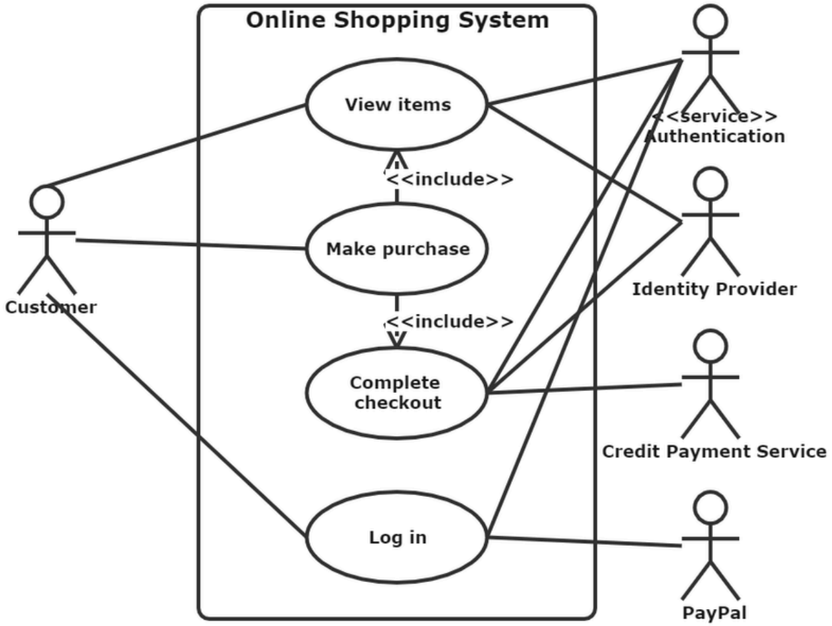


Application Layer

Application Layer is the layer that connects interface layer to the domain layer.

here there are many usecases of business present that may use more that one domain method for its working. along with that it is responsible for authentication and transaction processing . sending notifications or triggering events.

when interface later recieves a resuqest it sends to appropriate usecase method in application layer that may contain multiple domain methods and triggers events accordingly.



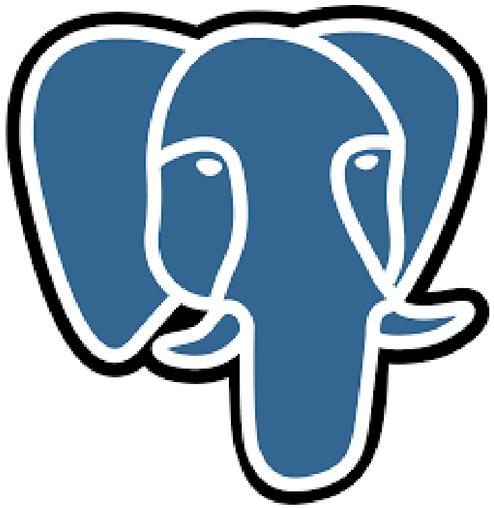
Interface Layer

Interface layer is the outer layer from where the applicaiton is accessed .
it can be a cli , a rest api or a grpc endpoint. even all three in one application .

This takes the appropriate input and sends to the application layer usecase for further processing and sends response back.

Infrastructure Layer

Infrastructure layer is where all interfaces are implemented concrete. It has all the technologies in it such as database, message queue, external apis and other relevent external things that are required by the application.



It should implements domain interface methods to work and this infra related technologies are injected into the repository or other related objects in the main setup. this process is called wiring.

Project

Github Link

https://github.com/subpxl/ecommerce-microservices

Github.com/subpxl

Shubham Panchal